PROJECT REPORT

ON

CAMPUS VEHICLE MANAGEMENT SYSTEM

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Course

Python (AI & ML)

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1. INTRODUCTION

Python is a high-level and general-purpose programming language known for its simple syntax and readability. The Campus Vehicle Management System is a Python-based application developed to demonstrate the practical implementation of Python fundamentals.

The system provides a simple way to manage vehicles, users, vehicle bookings, trips, and reports within a college campus. Instead of maintaining transportation information manually, the application organizes the information through a menu-driven program.

The project demonstrates the use of Python concepts such as variables, input/output operations, operators, conditional statements, loops, functions, lists, dictionaries, validation, and error handling to develop a practical application.

2. PROBLEM STATEMENT

Colleges and universities may use multiple vehicles for transportation and official campus activities. Managing vehicle details, users, bookings, and trip records manually can be time-consuming and difficult. Manual records may also make it harder to search for information and maintain organized data.

The Campus Vehicle Management System provides a simple computerized solution using Python. It allows vehicle, user, booking, and trip information to be managed through separate functional modules and provides reports for viewing stored information.

3. FUNCTIONAL REQUIREMENTS

The Campus Vehicle Management System consists of five major functional modules:

Vehicle Management: Allows the user to add, view, search, and delete vehicle records.

User Management: Allows the user to add, view, search, and delete user records.

Booking Management: Allows vehicle bookings to be created, viewed, searched, and deleted.

Trip Management: Allows trip information to be added, viewed, searched, and deleted.

Reports: Displays vehicle, user, booking, and trip information and provides an overall system summary.

4. NON-FUNCTIONAL REQUIREMENTS

Usability: The system provides a simple menu-driven interface that is easy to understand and operate.

Performance: Basic operations such as adding, viewing, searching, and deleting records are performed quickly.

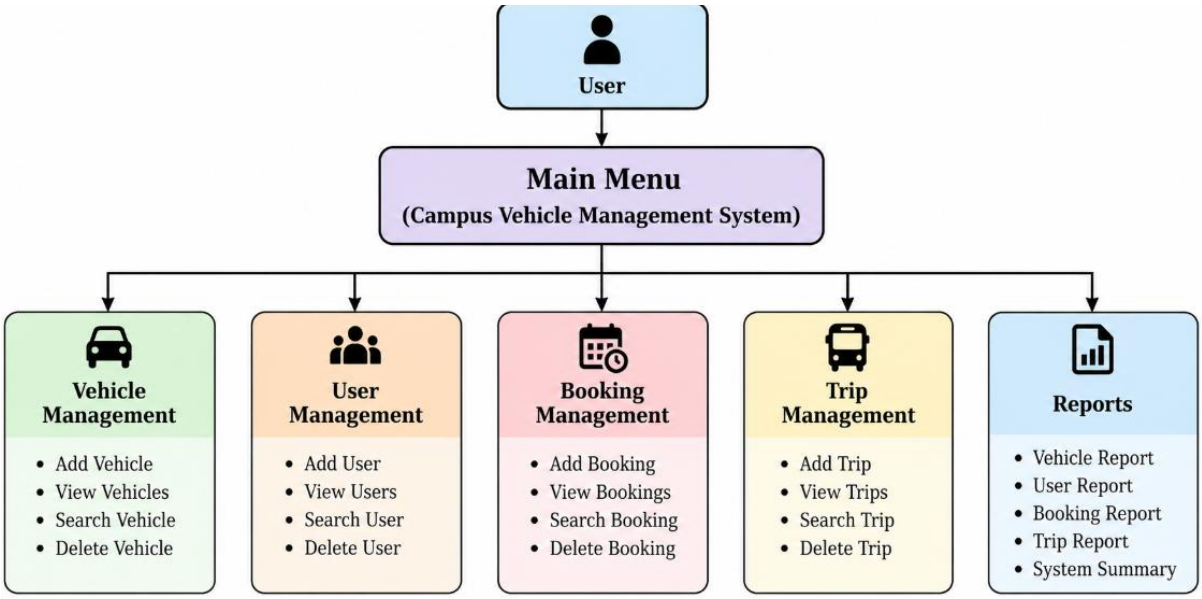
Reliability: The program provides appropriate messages for invalid choices and when requested records are not found.

Maintainability: The program is divided into separate functions and modules, making the code easier to understand and modify.

Error Handling: Invalid menu selections are handled properly without terminating the entire program.

5. SYSTEM ARCHITECTURE

The Campus Vehicle Management System follows a modular architecture. The Main Menu acts as the central controller of the application and provides access to Vehicle Management, User Management, Booking Management, Trip Management, and Reports. Each module performs its own specific operations while remaining connected through the Main Menu.



6. DESIGN DIAGRAMS

6.1 Workflow Diagram

The system starts by displaying the Main Menu. The user selects Vehicle Management, User Management, Booking Management, Trip Management, or Reports. After completing the selected operation, the user can return to the Main Menu. The program continues until Exit is selected.

6.2 Use Case Diagram

The main actor of the system is the Administrator. The Administrator can manage vehicles, manage users, manage bookings, manage trips, and view reports.

6.3 Sequence Diagram

The user selects an option from the Main Menu. The Main Menu calls the corresponding management function. The selected module accepts user input, processes the request, stores or retrieves the required information, and displays the result.

6.4 Component Diagram

The system consists of five major components: Vehicle Management, User Management, Booking Management, Trip Management, and Reports. These components are controlled through the Main Menu.

7. DESIGN DECISIONS AND RATIONALE

A menu-driven approach was selected because it provides a simple and easy-to-use interface. Lists and dictionaries are used to organize records during program execution. Functions are used to separate the major operations of the system, improving readability and maintainability.

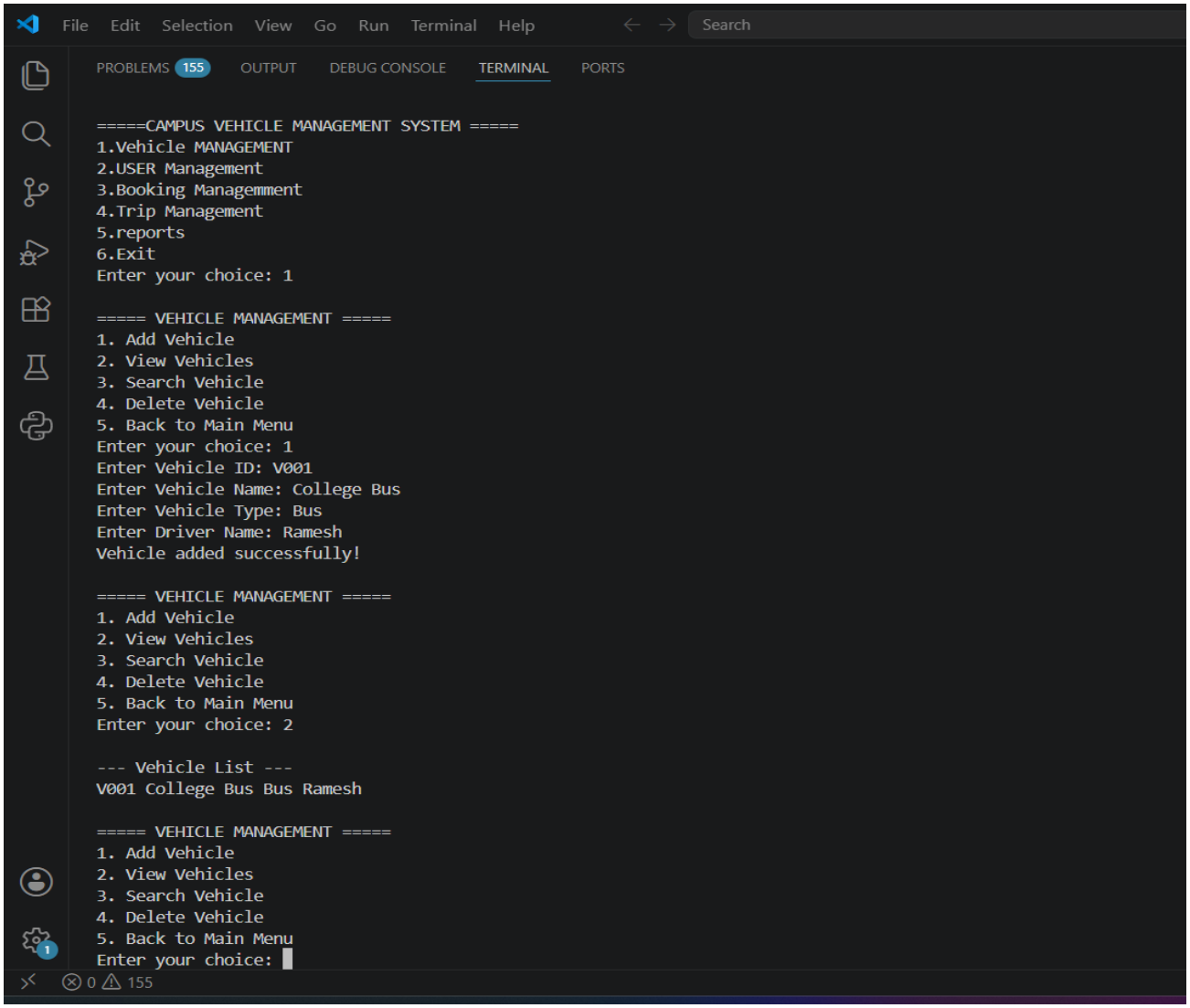
8. IMPLEMENTATION DETAILS

The project was implemented using Python in Visual Studio Code. The application uses input/output operations, conditional statements, loops, functions, lists, dictionaries, operators, and error handling.

The Main Menu provides access to Vehicle Management, User Management, Booking Management, Trip Management, and Reports. Each module contains options for performing its respective operations.

9. SCREENSHOTS AND RESULTS

The Campus Vehicle Management System was successfully executed and tested. The system displayed the Main Menu correctly and successfully performed vehicle addition and viewing operations.



10. TESTING APPROACH

The program was tested by executing the major modules individually. Vehicle records were added, viewed, and searched. User, booking, and trip modules were also checked. Invalid menu choices were tested to verify error handling.

The system successfully displayed an error message for invalid input without terminating unexpectedly.

11. CHALLENGES FACED

During development, challenges were faced with Python indentation, nested loops, function calls, and menu navigation. Some errors occurred while connecting individual management functions to the Main Menu. These problems were solved through debugging and repeated testing.

12. LEARNINGS AND KEY TAKEAWAYS

This project provided practical understanding of Python syntax, variables, input/output, operators, conditional statements, loops, functions, lists, dictionaries, searching, validation, error handling, and modular programming.

The project also provided experience in debugging programs and developing a complete menu-driven Python application.

13. FUTURE ENHANCEMENTS

The system can be improved by adding permanent database storage, login authentication, a graphical user interface, automatic vehicle availability checking, prevention of duplicate IDs, booking dates and time slots, and GPS-based vehicle tracking.

14. CONCLUSION

The Campus Vehicle Management System demonstrates the practical application of Python fundamentals in solving a campus transportation management problem. The system provides separate modules for vehicles, users, bookings, trips, and reports.

The project successfully demonstrates how fundamental Python concepts can be combined to develop an organized and functional application.

15. REFERENCES

1. Python Software Foundation – Python Documentation

2. Visual Studio Code Documentation

3. VIT Bhopal University course materials and classroom notes